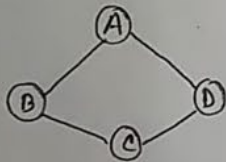


Minimum Spanning Tree

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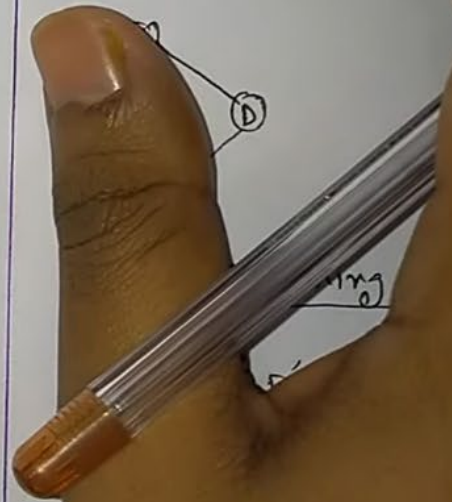
- ⇒
- (i) subset of a graph.
 - (ii) cycle হার না।
 - (iii) Vertices disconnected হার না।
 - (iv) নতুন কোন edge add/delete করা যায় না।



$G(V, E)$

$S(V', E')$

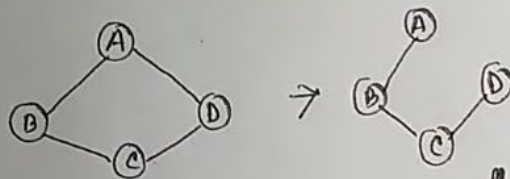
$|E| = |V| - 1$



- ⇒
- (i) subset of a graph.
 - (ii) cycle होना नहीं।
 - (iii) Vertices disconnected होना नहीं।
 - (iv) नया edge add/delete कर सकते हैं।

04

0-0



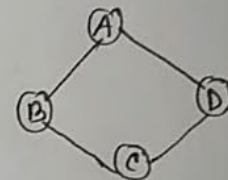
Spanning Tree

$G(V, E)$

$S(V', E')$

$V' = V$
 $E' = |V| - 1$

Complete graph



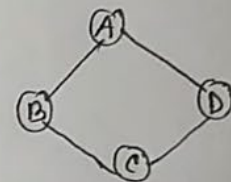
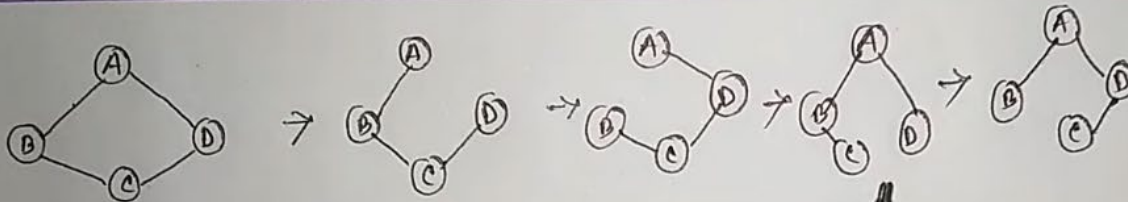
No. of spanning trees
 $\frac{E!}{E'!} - \text{no. of}$

No. of MST
 $\frac{V^{V-2}}$

edge
 $V + 1$

- ⇒
- (i) Subnet of a graph.
 - (ii) cycle શરૂ ના.
 - (iii) Vertices disconnected શરૂ ના.
 - (iv) નવું જોડાણ edge add/delete કરવા શરૂ ના.

04



$G(V, E)$

$S(V', E')$

$V' = V$
 $E' = E - 1$

$E \rightarrow 4$
 $E' \rightarrow 3$
 $E' \rightarrow 4$

Number of Spanning Tree we can get from this graph

No. of spanning tree :
 $E - E' - \text{no. of ex}$

complete graph

formula



MST
 $V - 2$

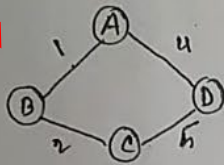
MST

Weighted graph

- ⇒
- (i) subset of a graph.
 - (ii) cycle હોવ ના.
 - (iii) Vertices disconnected હોવ ના.
 - (iv) નવું જોડાણ edge add/delete કરવા પાડે ના.

04

0 — 0

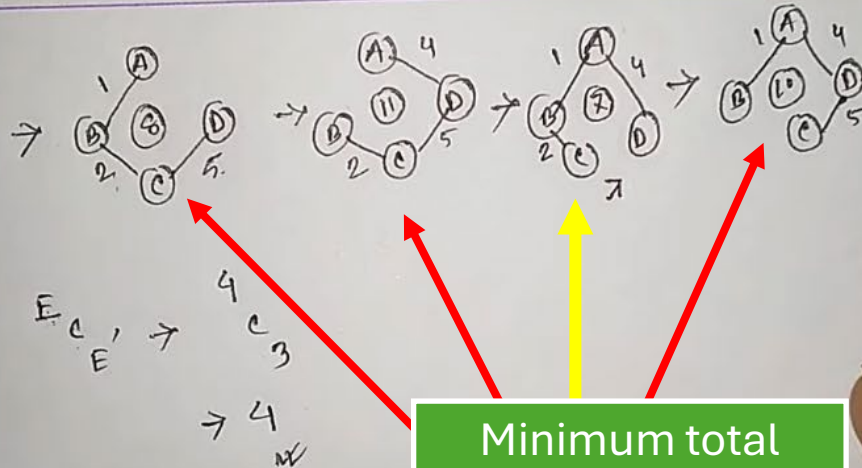


$G(V, E)$

$S(V', E')$

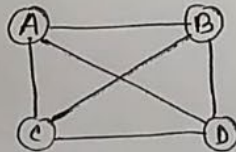
$|V|-1$

complete graph



Minimum total weight among the possible weighted spanning trees

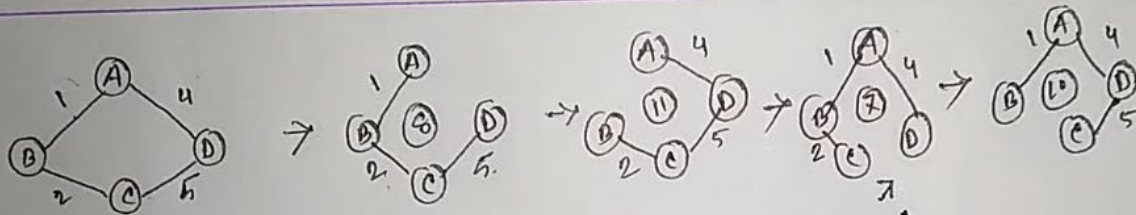
$$E = V + 1$$



- ⇒
- (i) Subnet of a graph.
 - (ii) cycle हाव ना।
 - (iii) Vertices disconnected हाव ना।
 - (iv) नएन एन edge add/delete कर हाव ना।

04

0 — 0

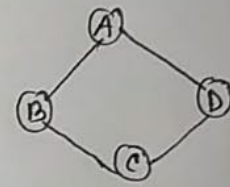
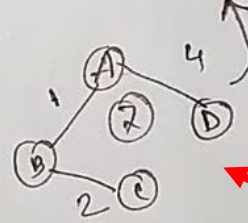


$G(V, E)$

$S(V', E')$

$$\begin{aligned} V' &= V \\ E' &= |V| - 1 \end{aligned}$$

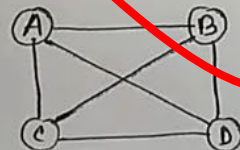
$E \subset E' \rightarrow 4 \subset 3$
 $\rightarrow 4$ ✗



No of spanning tree :
 No of e

Just take the minimum weighted edges from the Weighted graph

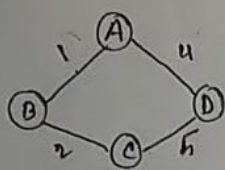
complete graph



MST

- (ii) cycle हाव ना।
- (iii) Vertices disconnected
- (iv) नएन एज edge add

Changes in the formula when there is an edge that creates cycle within the graph

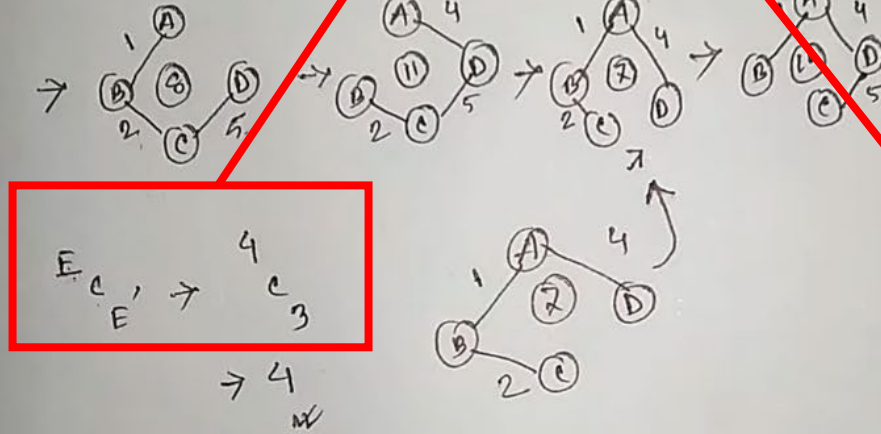


$$G(V, E)$$

$$S(V', E')$$

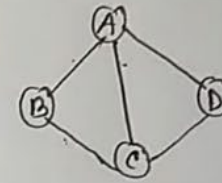
$$V' = V$$

$$E' = |V| + 1$$



$$E_{c, E'} \rightarrow 4, 3$$

$$\rightarrow 4$$

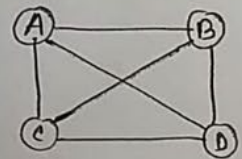


$$\text{No of spanning tree:}$$

$$E_{c, E'} - \text{no of cycle}$$

$$5 - 2 = 3$$

Complete graph



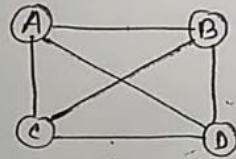
Remove edge

$$E = V + 1$$

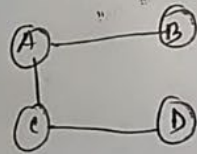
No of MST

$$V^{V-2}$$

Complete graph



↓



Remove edge

$$E = V + 1$$

$$\Rightarrow 6 = 4 + 1$$

$$\Rightarrow 2 + 1$$

$$\Rightarrow 3$$

No of MST

$$V^{V-2}$$

$$4^{4-2}$$

$$\Rightarrow 4^2$$

$$= 4$$

$$= 16$$

